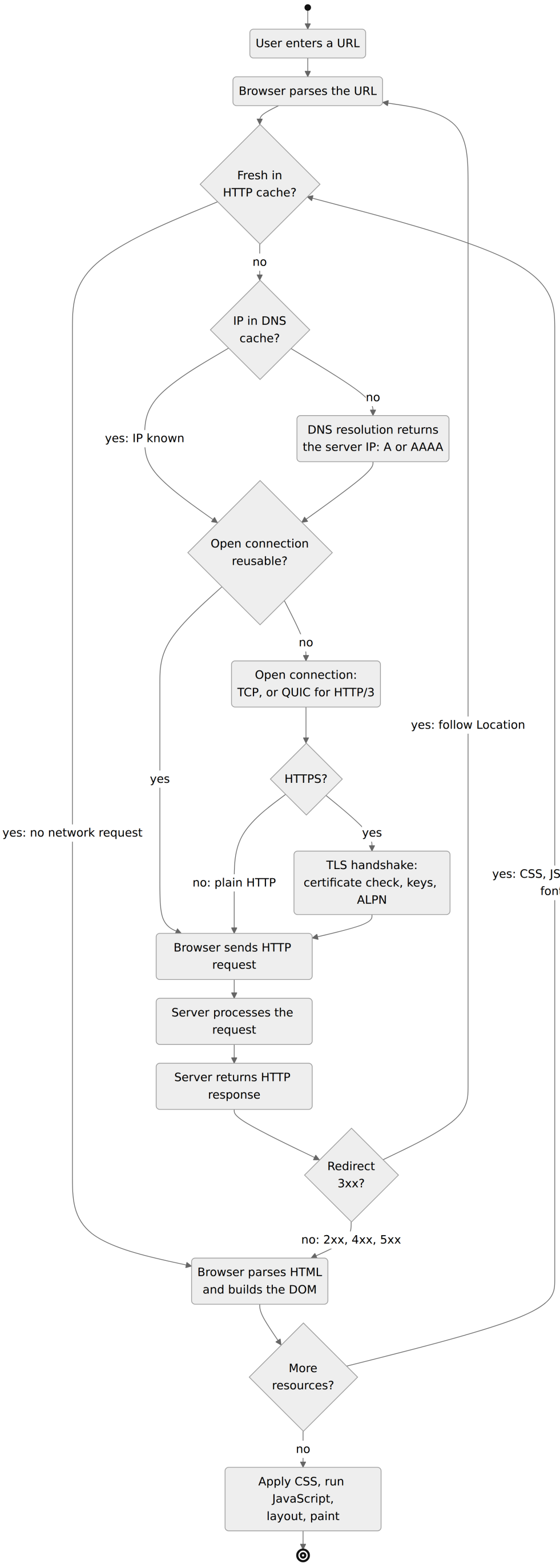
What Happens When a URL Is Entered in a Browser

High-level UML activity diagram | Week 1 web development internship | Companion to Section A of week-1-combined-research.pdf



Legend

- Start (initial node)
- End (final node)
- Action (a step the browser or server performs)
- Decision (branches are labelled)
- Control flow

Reading notes

Not every request takes every step. A fresh cache hit skips the network entirely. A cached IP skips DNS. An open connection skips the TCP and TLS setup.

Cached IP. The "IP in DNS cache?" check covers the browser cache and the operating system cache. If both miss, the OS asks a DNS resolver, which may already have the answer cached too.

HTTP/3 differs. HTTP/3 runs over QUIC (on UDP) and builds TLS 1.3 into the connection handshake, so the separate TCP and TLS boxes are merged into one step in that case.

ALPN. During the TLS handshake the client and server agree on the application protocol, for example HTTP/2 or HTTP/1.1.

HTTP to HTTPS. A request to an http:// address is often answered with a redirect to https://. The 3xx loop back to "Browser parses the URL" shows this case. Browsers can also upgrade known hosts directly (HSTS).

Error responses. A 404 or 500 is still a complete HTTP response. The browser renders the error page it receives.

Resource loop. The HTML is only the first response. Each stylesheet, script, image and font found in it goes back through the cache check, and usually reuses the open connection.

Notation. Mermaid has no dedicated UML activity diagram type. This is a Mermaid flowchart that follows activity diagram conventions. Fork and join bars for parallel downloads are left out to keep the diagram readable.

Editable source: browser-request-flow.mmd (Mermaid)